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Geology 1982;10;423-428
doi: 10.1130/0091-7613(1982)10<423:CMINNA>2.0.CO;2

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Cretaceous magnetizations in northwestern Nevada and tectonic implications

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ABSTRACT

Paleozoic and Mesozoic arc-related rocks of the Black Rock Desert region of northwestern Nevada have acquired a very stable postfolding secondary magnetization, probably in Late Cretaceous time. The remagnetization, which appears to be of regional extent, probably accompanied emplacement of granitic plutons in a northeast-trending belt across northwestern Nevada near the end of Mesozoic time. The region has experienced no appreciable tectonic rotation since that time.

INTRODUCTION

The Black Rock Desert region (Russell, 1981) contains Paleozoic and Mesozoic arc-related rocks that record a complex history of volcanism, sedimentation, and deformation before emplacement of Cretaceous (80 to 100 m.y. ago) batholithic rocks (Kistler and others, 1971; Smith and others, 1971). Nearby provinces of pre-Cretaceous age in the northern Sierra Nevada, eastern Klamath Mountains, and eastern Oregon (Fig. 1) are generally comparable to the Black Rock region in age range and arc affiliation of the rocks they contain. However, the extensive Cenozoic cover between the four provinces makes tectonostratigraphic correlations uncertain, and it remains to be established whether the provinces are exposures of a contiguous terrane (Speed, 1979) or consist of several unrelated terranes.

Paleomagnetic studies show differences in the Cenozoic histories of the four provinces. Magnetizations acquired in Late Jurassic or Early Cretaceous time in the northern Sierra province (Hannah and Verosub, 1980) and in Cretaceous time in the Black Rock Desert (this paper) are concordant, and the rocks thus are apparently unrotated (with respect to stable North America). In contrast, Jurassic plutons of the eastern Oregon inlier underwent 45° ± 36° clockwise rotation during or prior to the Eocene (Wilson and Cox, 1980), followed by late Eocene or younger clockwise rotation of about 15° (Beck and others, 1978).

No significant shift in latitude accompanied the rotation. Similarly, large clockwise rotations are evident in Eocene Tyee-Flournoy strata of the Oregon Coast Range, which lie unconformably on Mesozoic rocks of the Klamath province (Simpson and Cox, 1977). The implication is that the Klamath province also may have rotated substantially during or after Eocene time. The contrasting directions of late Mesozoic magnetizations among the four provinces indicate that an important tectonic boundary exists between them (Fig. 1). The boundary has no evident expression in the largely Neogene volcanic cover.

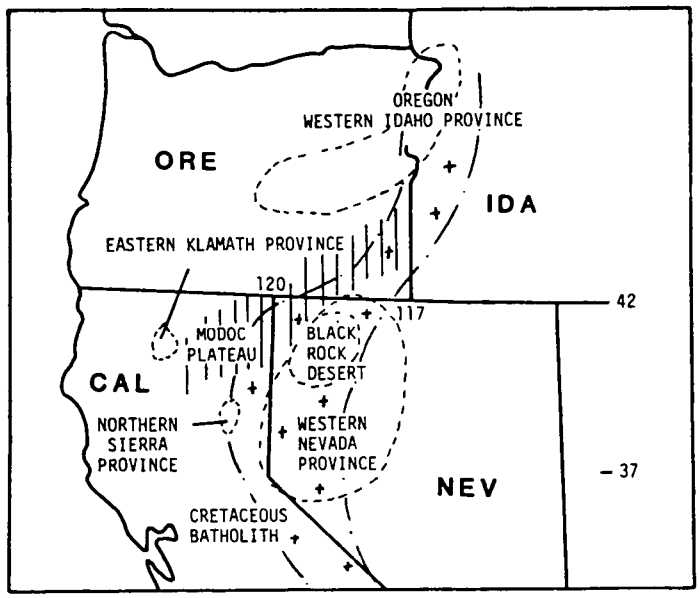


Figure 1. Selected tectonic provinces of central-western margin of North American Cordillera.

BLACK ROCK DESERT: TECTONOSTRATIGRAPHY

We have studied pre-Tertiary rocks of the Jackson Mountains and the Bilk Creek Range (Fig. 2). These ranges expose thrust complexes of Paleozoic (chiefly Permian) and Mesozoic rocks. It is not clear if these allochthonous units are far traveled with respect to their present sites. If not, they record the development of a magmatic arc in latest Triassic and Jurassic time on a deeply subsided Paleozoic basement.

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Numerous plutons that cut the aforementioned units probably are exposures of a Cretaceous batholith that extends from the Sierra Nevada to central Idaho and beyond (Kistler and others, 1971; Smith and others, 1971). Seven plutons within 50 km of the Black Rock Desert have been dated by biotite K/Ar as between 90 and 97 m.y. old (Smith and others, 1971). Except for one pluton, hornblende and biotite dates are concordant.

Our paleomagnetic sampling in the Black Rock region was confined to four localities, three of Jurassic and Cretaceous rocks in the Jackson Mountains and one of Permian rocks in the Bilk Creek Range (Fig. 2). The tectonostratigraphic setting of the sites is outlined below.

Jackson Mountains

Figure 2 shows a simplified tectonostratigraphic map in which thrust packets are grouped into two rock units, the McGill Canyon succession and the Jackson Mountains succession. The McGill Canyon succession is structurally higher and consists of upper Paleozoic volcanogenic and terrigenous turbidites and affiliated strata, with massive carbonate blocks of shoal-water origin. The Jackson Mountains succession contains intermediate and lower packets of Middle and Upper Triassic starved basin and slope strata, locally with carbonate olistostromes and/or volcanogenic turbidites. The rocks we sampled are restricted to the Jackson Mountains succession and are as follows: (1) the Happy Creek igneous complex of early Mesozoic age; and (2) the King Lear Formation, of Early Cretaceous age, which uncomformably overlies the Happy Creek (Willden, 1958) and other rocks of the Jackson Mountains succession. The Happy Creek igneous complex consists of andesite and related intrusive rocks that evolved from multiple, fractionated magmas of island-arc affinity (Russell, 1981). A four-point Rb/Sr isochron yields an age of 169 ± 5 m.y. for the Happy Creek complex (analyses by R. W. Kistler; Russell, 1981). The King Lear Formation is composed of alluvial volcano-

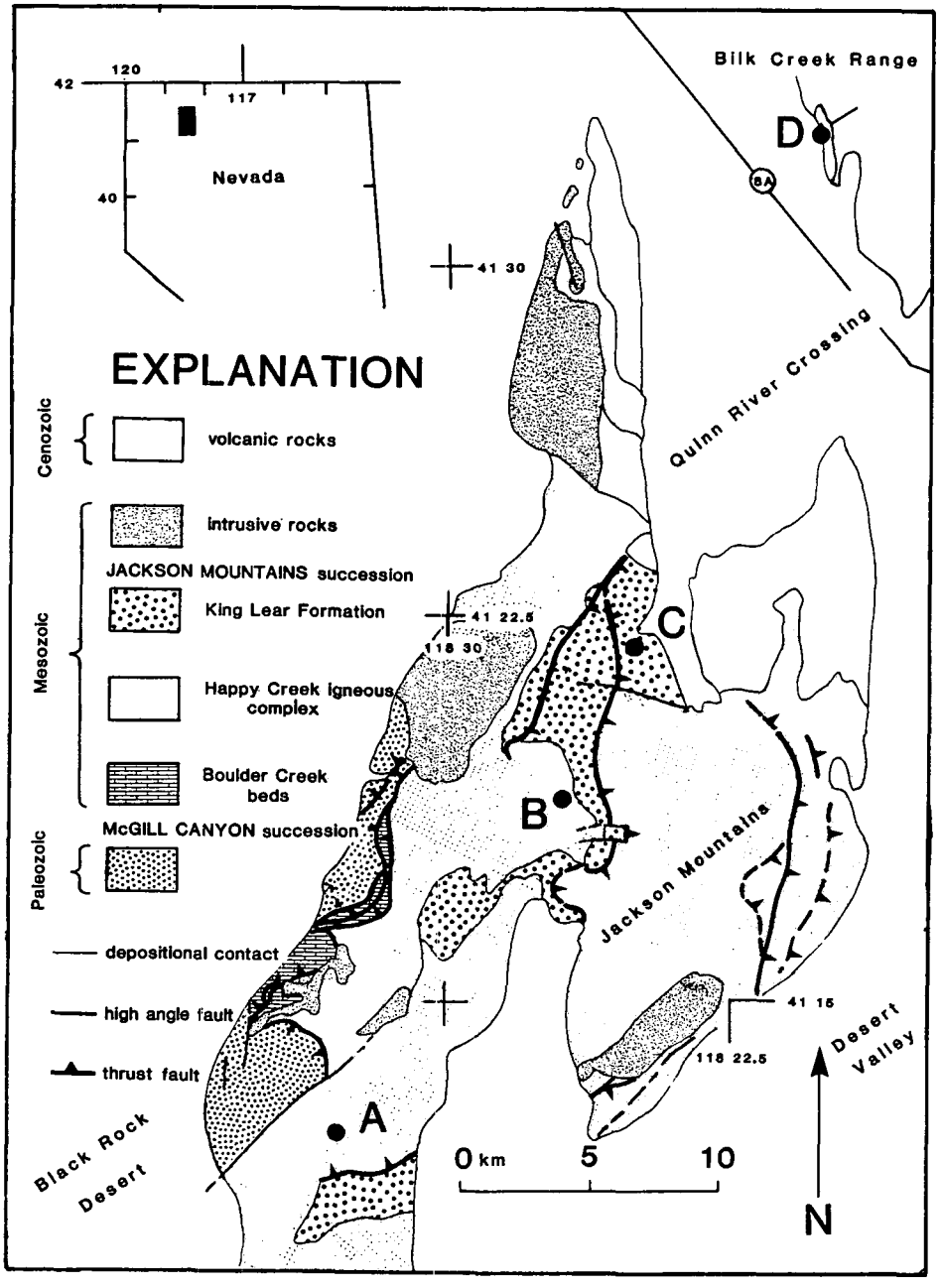


Figure 2. Simplified tectonostratigraphic map of sampled region, from Russell (1981). A–D are sampling localities, discussed in text.

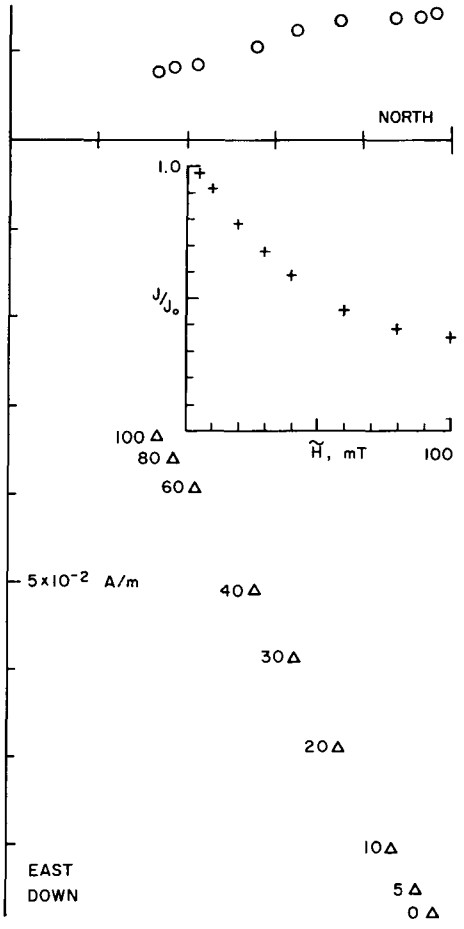


Figure 3. Typical results of AF demagnetization of King Lear rocks.

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genic conglomerate, finer grained red beds, and lacustrine limestone that accumulated above the Happy Creek in tectonic basins.

During or following emplacement of the Happy Creek complex, an episode of deformation resulted in northwest-striking, open, multikilometre-wavelength folds. Later, a second phase of deformation, orthogonal to the first, produced major and minor folds of northwest vergence and an axial-plane cleavage during southeast-northwest shortening. The Lower Cretaceous King Lear Formation was deposited and deformed during the second phase. Tectonic stacking occurred during and following the second phase of folding, after deposition of the King Lear Formation. The absence of significant metamorphism and the modest foliation indicate that thrusting was thin skinned.

Bilk Creek Range

Pre-Tertiary rocks of the Bilk Creek Range crop out near the point labeled D in Figure 2. Lower Permian limestones at Quinn River Crossing (Willden, 1964) were sampled for paleomagnetic measurements. Rocks at Quinn River Crossing record two phases of deformation, of which the second is comparable to the second phase in the Jackson Mountains.

PALEOMAGNETISM

Samples were collected from 17 individual flows at two widely spaced localities (A and B, Fig. 2) in the Happy Creek igneous complex and from 18 sites in the King Lear Formation. At locality B, dips are steep and southeasterly, whereas at A, beds dip more gently, to the west. This divergence permitted a useful fold test, which (as shown below) failed dramatically. Fine-grained red beds of the King Lear Formation were sampled in a single folded sequence (loc. C, Fig. 2) that dips 20° to 50° south, southwest, and southeast. A King Lear "site" consisted of samples from several (usually at least three) stratigraphic horizons over lateral distances of as much as 100 m. In both the Happy Creek complex and the King Lear Formation, a site consisted of five or more separately oriented samples.

Samples were drilled and oriented in situ by magnetic and solar readings, and the solar readings were reduced on the spot.

TABLE 1. DIRECTIONS OF REMANENT MAGNETIZATION IN THE HAPPY CREEK VOLCANICS, JACKSON MOUNTAINS, NEVADA

	In situ		After tectonic correction				
Site	Dec	Inc	Dec	Inc	N	R	H
HC 1*	323.5	-4.0	327.0	50.5	4	3.856	0
2*	316.0	40.0	55.5	67.0	10	8.707	0
3	312.5	13.5	347.5	64.5	7	6.845	100
4	304.5	4.0	320.5	62.0	7	6.985	100
5	308.0	7.5	338.0	63.0	6	5.911	80
6	340.0	57.0	71.5	48.5	6	5.915	20
7	317.0	41.0	51.0	64.0	7	6.833	30
8	334.0	59.5	75.0	49.5	5	4.979	25
20	346.0	63.0	297.5	35.0	6	5.990	20
21*	341.0	47.5	310.0	24.0	4	3.849	25
22*	319.0	65.0	287.0	27.0	4	3.965	25
23	206.5	-63.5	120.5	-53.5	6	5.976	0
24	327.0	61.5	290.0	28.5	6	5.959	60
25	335.0	65.5	288.5	34.0	11	10.940	25
26	357.0	9.0	346.5	14.5	7	6.916	100
27*	355.5	49.0	308.5	36.5	2	1.987	100
28	349.5	66.5	286.0	39.5	4	3.990	30

Note: Dec = declination; Inc = inclination; N = number of samples; R = length of vector sum; H = peak value of AF demagnetizing field, in mT.

*Indicates site for which α_{95} exceeds 15°.

TABLE 2. DIRECTIONS OF NATURAL REMANENT MAGNETIZATION IN THE KING LEAR FORMATION, JACKSON MOUNTAINS, NEVADA

	In situ				After tectonic correction			
Site	Dec	Inc	N	R	Dec	Inc	N	R
KL01	2.0	68.0	7	6.963	149.0	65.0	7	6.920
02*	346.5	68.0	6	5.943	156.0	76.0	6	5.565
03*	348.5	63.5	2	1.980	36.0	86.0	2	1.980
04	342.5	69.0	9	8.788	154.0	66.0	9	8.825
05	341.0	64.0	7	6.944	177.5	77.5	7	6.900
06	349.5	61.0	7	6.704	156.0	75.0	7	6.786
07	292.0	67.0	6	5.925	215.0	46.5	6	5.876
08	332.5	61.5	6	5.918	227.0	43.5	6	5.858
09*	19.0	72.7	7	6.478	270.0	38.5	7	6.123
10	308.5	77.5	10	9.326	264.0	40.0	10	9.410
11	60.0	65.0	6	5.794	242.0	71.5	6	5.882
12	19.0	75.1	7	6.932	213.0	24.0	7	6.978
13	330.5	54.4	6	5.941	286.0	52.0	6	4.908
14	337.5	60.5	7	6.966	196.5	70.0	7	6.970
15	339.5	65.0	7	6.985	174.0	59.0	7	6.979
16	343.0	60.0	7	6.940	171.5	66.0	7	6.933
17	334.5	61.5	8	7.986	180.0	56.0	8	7.990
18	351.0	64.0	7	6.981	154.0	61.5	7	6.981

Note: Dec = declination; Inc = inclination; N = number of samples; R = length of vector sum.

*Indicates site for which α_{95} exceeds 15°.

This greatly minimized errors due to misorientation and to abnormally strong magnetizations caused by lightning. Our orienting technique is accurate to 2° or less.

All samples, including sedimentary rocks, had remanent magnetizations that could be measured easily on a Schonstedt SM-1a spinner magnetometer. Pilot alternating-field (AF) demagnetization runs to peak fields of 100 mT were made on selected specimens from each site, and additional samples were cleaned thermally to 650 °C or more. In general, the Happy Creek volcanic sites responded well to AF magnetic cleaning, and all but three sites were cleaned by this method, in peak fields ranging from 25 to 100 mT (Table 1). The three sites reported at normal remanent magnetization (NRM) (0 mT) in Table 1 all failed to respond to either form of magnetic cleaning; two of these are highly scattered at NRM, but one, site HC23, is tightly clustered.

AF magnetic cleaning proved useless for King Lear rocks. At least two test specimens from each site were demagnetized to peak fields of 100 mT; Figure 3 shows a typical result and demonstrates that the magnetic directions are quite stable. Thermal demagnetization to temperatures as high as 657 °C on pilot samples likewise failed to produce significant, consistent changes in magnetic direction. We conclude that King Lear rocks have a stable magnetization that, moreover, is secondary (as shown below). Because magnetic cleaning failed to reduce scatter or change mean directions to any appreciable degree, in Table 2 we present NRM results for King Lear rocks. Most King Lear sites cluster very well, confirming other indications that its NRM is extremely stable.

The magnetization of the Happy Creek and King Lear rocks at localities A–C (Fig. 2) is secondary and postfolding. Figure 4a shows expected directions (ED) for Mesozoic and Cenozoic remanent magnetization in the Black Rock region, calculated from reference poles tabulated by Irving (1979) on the assumption that the region has always been part of the North American craton. Barring rock-magnetic complications, rocks in the Jackson Mountains should show approximately these directions unless the rock body has moved since it acquired its magnetization.

Figure 4b summarizes the results for the lower King Lear beds, which dip moderately to the south, southeast, or southwest.

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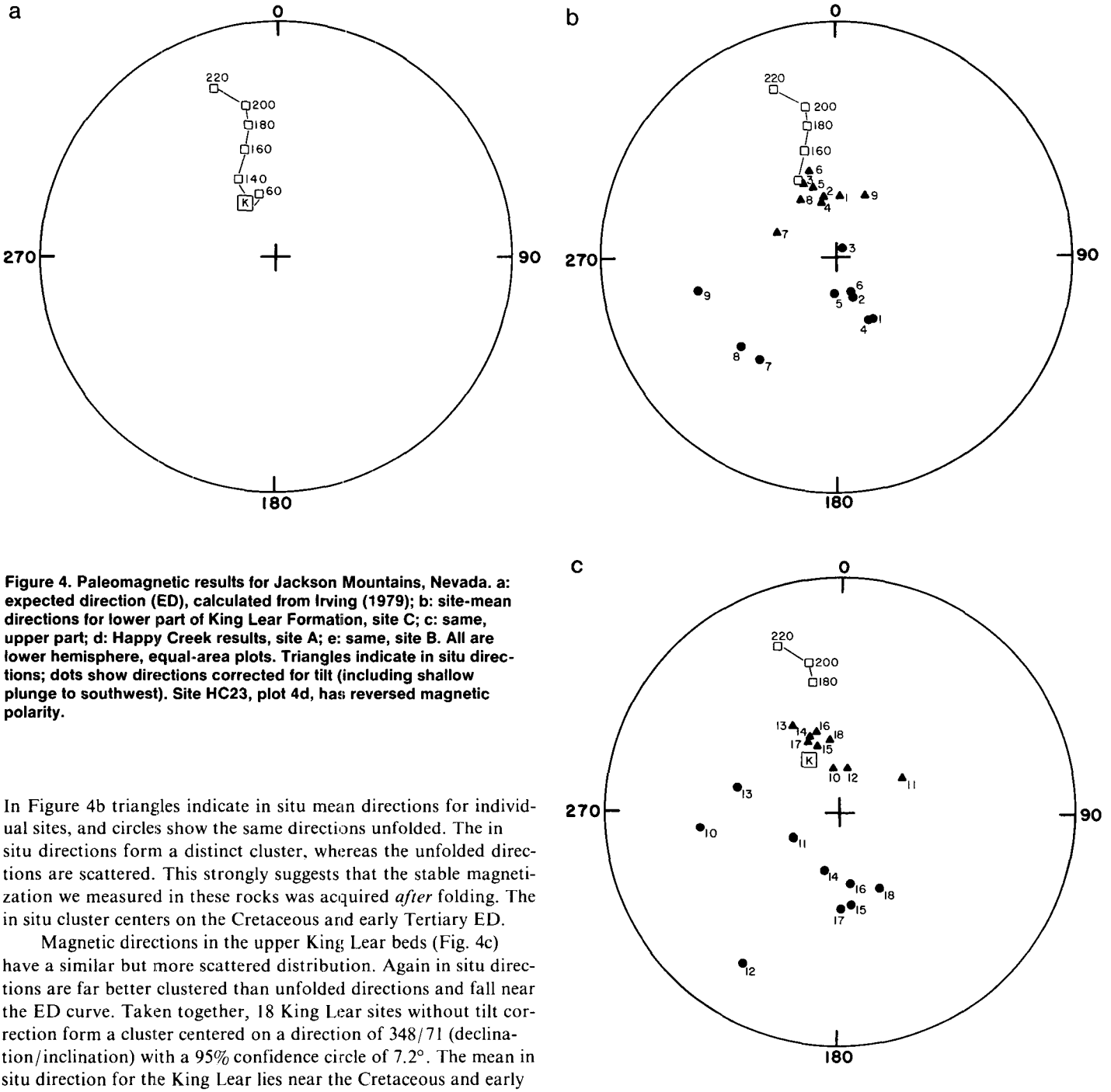


Figure 4. Paleomagnetic results for Jackson Mountains, Nevada. a: expected direction (ED), calculated from Irving (1979); b: site-mean directions for lower part of King Lear Formation, site C; c: same, upper part; d: Happy Creek results, site A; e: same, site B. All are lower hemisphere, equal-area plots. Triangles indicate in situ directions; dots show directions corrected for tilt (including shallow plunge to southwest). Site HC23, plot 4d, has reversed magnetic polarity.

In Figure 4b triangles indicate in situ mean directions for individual sites, and circles show the same directions unfolded. The in situ directions form a distinct cluster, whereas the unfolded directions are scattered. This strongly suggests that the stable magnetization we measured in these rocks was acquired *after* folding. The in situ cluster centers on the Cretaceous and early Tertiary ED.

Magnetic directions in the upper King Lear beds (Fig. 4c) have a similar but more scattered distribution. Again in situ directions are far better clustered than unfolded directions and fall near the ED curve. Taken together, 18 King Lear sites without tilt correction form a cluster centered on a direction of 348/71 (declination/inclination) with a 95% confidence circle of 7.2°. The mean in situ direction for the King Lear lies near the Cretaceous and early Tertiary EDs for the North American craton.

Magnetic directions of the Happy Creek complex are more scattered than those of the King Lear Formation. Figure 4d shows results for locality A (Fig. 2). Directions for homoclinal flows at sites 20, 22, 24, 25, and 28 without unfolding form a tight cluster centered on the Cretaceous ED; unfolded directions for the same sites also cluster, with a strong westerly declination and flattened inclination. From these sites alone one might speculate that the Jackson Mountains had been translated from far to the south and rotated counterclockwise since the eruption of the Happy Creek complex. However, the close agreement of the in situ cluster with remagnetized directions of King Lear rocks suggests instead that these Happy Creek flows also were thoroughly remagnetized after folding. Sites 21, 26, and 27 give mean directions outside any cluster, for which we have no explanation. Site 23, which also has a peculiar direction, is interesting in that it has reverse magnetic polarity. Sites 21 and 22 also have two samples, each with reverse polarity (the remaining samples from each site are normal); these fall very near the directions for site 23. Perhaps remagnetization of the Happy Creek rocks spanned a polarity transition. The fact that the unfolded direction for the reverse Happy Creek magnetizations lies midway between the unfolded and in situ clusters for sites of normal polarity (Fig. 4d) may be a coincidence. If not, it suggests that remagnetization of the reverse Happy Creek volcanics occurred during folding, or that the reversed rocks are incompletely remagnetized.

Figure 4e shows results for Happy Creek rocks at locality B (Fig. 2). As in the other three plots, there is a cluster centered on the ED curve. However, this cluster contains tilt-corrected site-mean directions (1, 3, 4, and 5) as well as in situ directions (6, 8).

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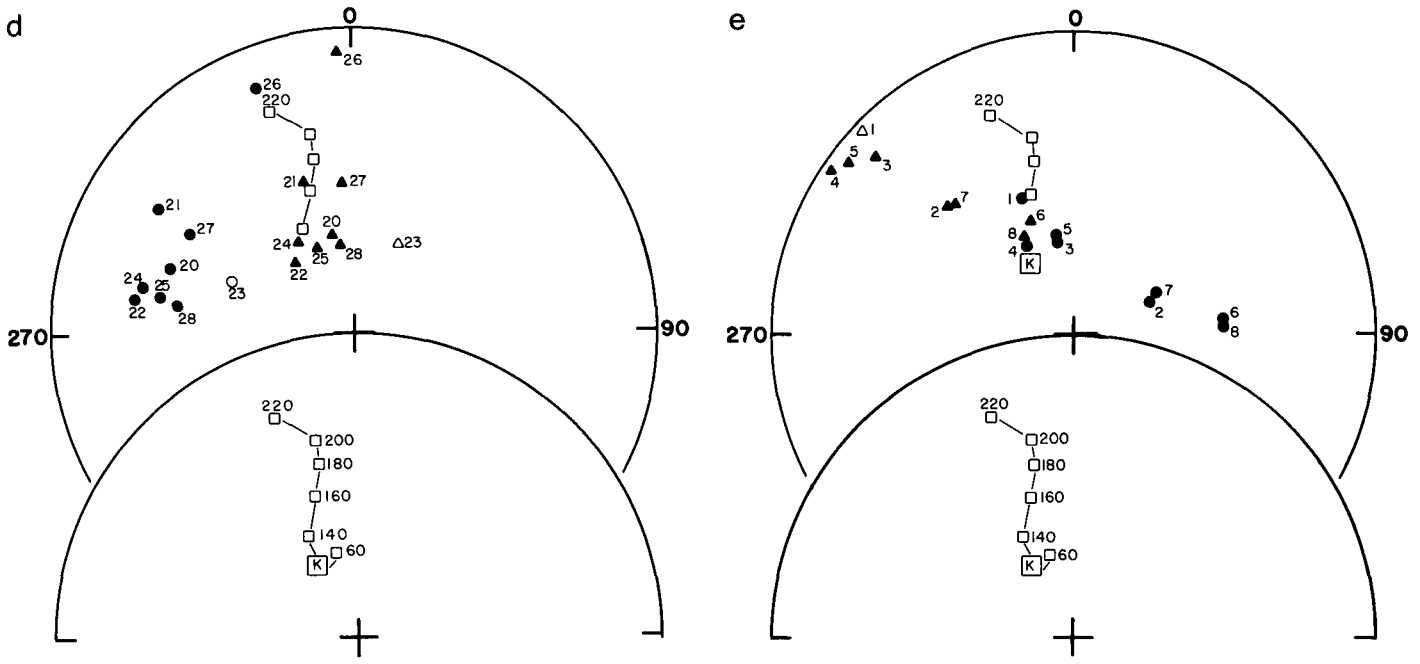


Figure 4. ***(Continued.)***

This suggests that remagnetization and folding occurred nearly simultaneously at locality B. This impression is enhanced by the scatter within sites 2 and 8, both of which have directions that form a linear distribution between the unfolded (tilt-corrected) direction and the main cluster.

From Figures 4b–4e it is hard to escape the conclusion that the Happy Creek and King Lear rocks were thoroughly remagnetized in late Mesozoic or very early Tertiary time.

We also collected 41 samples from the Lower Permian limestone of the Bilk Creek Range. Unfortunately, these samples are magnetized at or below the lower limit of sensitivity of our instrument. Results have been attained at NRM for 38 samples, and these cluster fairly well (R = 34.585) at a direction (declination, 326°; inclination, 73°) very near the Cretaceous ED. Tentatively, we suspect the Bilk Creek rocks also have been involved in the wholesale remagnetization.

INTERPRETATION

Remagnetization obviously postdates the Early Cretaceous deposition of the King Lear Formation and subsequent folding of these beds. The remagnetization was thus later than the second phase of deformation in the Jackson Mountains (Russell, 1981). However, because several Happy Creek flows apparently retain prefolding (1, 3, 4, 5) and intermediate (2, 7) magnetic directions, the remagnetization of flows at least at locality B (Fig. 2) seems to have been nearly contemporaneous with folding. The existence of remagnetized reverse-polarity rocks in the section suggests that remagnetization did not occur during the long normal-polarity interval extending (Lowrie and others, 1980) from the early Barremian through Santonian (about 120 to 80 m.y. B.P.). Very early Cretaceous or very late Cretaceous–early Tertiary seem to be the likeliest candidates.

The paleomagnetic data themselves can be used to discriminate between these alternative ages, although an uncomfortable degree of subjectivity is involved. On the assumption that King Lear and Happy Creek rocks were remagnetized at the same time and not disturbed since, we grouped "clustered" directions from both units (all in situ directions of the King Lear Formation and the following Happy Creek sites: 6, 8, 20, 21, 22, 24, 25, 27, 28, in situ; 3, 4, 5 after correction for dip). The remaining Happy Creek sites are excluded because they do not seem to belong to the main cluster, a criterion that admittedly is difficult to defend. From this data set we dropped all mean directions that are poorly determined (the cutoff level was set arbitrarily at a 95% confidence circle of 15°; such sites are marked with an asterisk in Tables 1 and 2). The resulting 24 site-mean directions should give a reliable estimate of the direction of the geomagnetic field at the time of remagnetization. The mean direction for these sites is 341.5/67 (D/I), with a 95% confidence interval of 5.0°, implying that Late Cretaceous to early Tertiary remagnetization was more likely. Figure 5, pole 6, shows this mean direction transformed to a

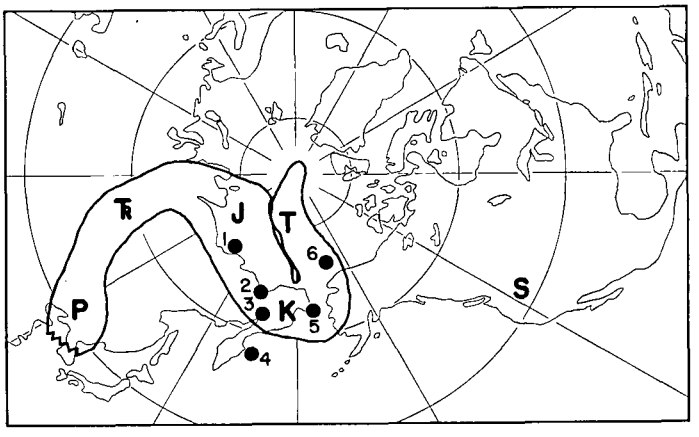


Figure 5. Pole positions for northern Sierra Nevada–northwestern Nevada area, plotting on polar-wandering path of Irving (1979). 1: Reeve Formation (Hannah and Verosub, 1980); 2: Taylor Formation (Hannah and Verosub, 1980); 3: Sierra Nevada Cretaceous granitic rocks (Grommé and Merrill, 1965); 4: Bucks batholith (Grommé and others, 1967); 5: Great Valley sedimentary rocks (Mankinen, 1978); 6: Jackson Mountains, Nevada (this paper). S indicates sampling site for Jackson Mountain study.

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paleomagnetic pole. It lies squarely between Cretaceous and early Tertiary positions (about 70 to 80 m.y. ago) on Irving's (1979) curve for North America.

Our data indicate that rocks of the Black Rock region have not rotated or changed latitude with respect to the North American craton since Late Cretaceous time. (Purely east-west movements, of course, cannot be excluded.) Evidently a major Cretaceous or early Tertiary tectonic boundary separates pre-Tertiary rocks of northwestern Nevada from rotated terranes in central and eastern Oregon (Wilson and Cox, 1980) and the Klamath Mountains of northern California and southwestern Oregon (Simpson and Cox, 1977). However, some Mesozoic rocks west of the Black Rock Desert region also yield paleomagnetic directions that parallel North American directions. This is illustrated in Figure 5, where paleomagnetic poles for the Sierra Nevada batholith (Grommé and Merrill, 1965; Grommé and others, 1967), for remagnetized country rock of the northern Sierra Nevada batholith (Hannah and Verosub, 1980), and for remagnetized Cretaceous sedimentary rocks from the Great Valley of California (Mankinen, 1978) are plotted on Irving's (1979) North American curve. All fall on or near the curve, and none is compatible with more than a few degrees of post-Cretaceous rotation. Wholesale Tertiary readjustment of crustal blocks in eastern California and Nevada, such as the scheme proposed by Magill and others (1981), thus seems unlikely in light of available paleomagnetic evidence.

Reviewer's comment

Sounds a useful note of caution in re-emphasizing the importance of the fold test, and the possibility of remagnetization, in paleomagnetic studies.

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ACKNOWLEDGMENTS

Reviewed by E. Mankinen and S. Levi. An earlier version was reviewed by K. Verosub. Supported by National Science Foundation Grant EAR 79-11150.

MANUSCRIPT RECEIVED JANUARY 4, 1982
REVISED MANUSCRIPT RECEIVED APRIL 7, 1982
MANUSCRIPT ACCEPTED APRIL 12, 1982